Submission Summary

Conference Name
IEEE Chandigarh Subsection International Conference

Track Name
Artificial Intelligence and Data Science

Paper ID
1115

Paper Title
ECG Arrhythmia Detection System Using CNN

Abstract
Cardiac arrhythmias are responsible for a large share of
cardiovascular deaths every year, and catching them early can
genuinely change patient outcomes. This paper describes a CNNbased system for automatic ECG arrhythmia classification built entirely on the PhysioNet MIT-BIH Arrhythmia Database—no custom hardware, no live signal acquisition, just a clean software pipeline. Raw recordings go through bandpass filtering, baseline removal, and beat-level segmentation before being fed into a fourblock convolutional network that predicts one of four rhythm categories:
Normal, Atrial Fibrillation (AF), Premature
Ventricular Contraction (PVC), or Other Arrhythmia. On the heldout test set the model reaches 94.7 % weighted accuracy and a macro AUC-ROC of 0.964, comfortably ahead of SVM and
random-forest baselines. GradCAM saliency overlays are included so that individual predictions can be checked against the actual waveform.

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3. Author's Information

I confirm that I have listed the title of the paper and the names, email addresses and affiliations of all contributed authors in the submission system that match the order as given in the submitted paper

Agreement accepted

4. Co-author's Consent

I confirm that I have discussed about this submission with all co-authors and authors have approved the paper and agree with its submission to this conference.

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5. Certificate of Originality

I confirm that neither the paper nor any parts of its content are currently under consideration or published in another conference/journal and the paper is free from plagiarism

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6. Author's Consent

It is confirmed that at least one author will register in full to attend and present the paper at the Conference.

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7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

I confirm that no text in the paper is AI-generated.

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